

Laboratory Session #1

Deliverables

This assignment is to be completed in pairs and will be submitted to Crowdmark (PDF). Pairing allocations are posted in the lab session and can also be found on LEARN.

In this assignment, you will:

- Write, debug, and run simple MatLab programs.
- Practice variable assignments and type-casting.
- Perform simple arithmetic computations using MatLab operators.
- Take input from the user.
- Provide simple outputs to the user.

Your Crowdmark submission should contain:

1. The code that you wrote.
2. The command window readout for your code, including input and output statements.
3. Your answers to the questions included in the assignment.

Crowdmark Submission with Partners

Prior to submitting your assignment work to Crowdmark, you need to tell Crowdmark that you are part of a group. To do this, complete the following steps:

- Navigate to the Lab Assignment submission area in Crowdmark.
- Click the “Add Group Members” button at Crowdmark and choose your group for this assignment. You can also wait for someone else to add you to their group. *Note that after the assignment has been submitted, your group cannot be changed.*
- Each group member you add will receive an email notification and shared access to this page.
- Any group member may submit the assignment or edit group members.

Part A

Problem Description

Write a program that prints out a restaurant receipt. This program must:

- Take input from the waiter for the cost of the meal.
- Take from the customer for the tip percentage.
- Calculate the sales tax, tip amount, and total bill.
- Output the total bill, including tax and tip.

Provide a 4-6 sentence description of the design decisions that went into the code, particularly in response to the questions outlined below.

Questions to Consider

As you write your code, please ponder the following questions. When you submit your code, please acknowledge any external references (e.g. websites) used in your design.

- Where is your restaurant located? How is sales tax assessed in this location?
- Restaurants typically either ask for tips as a dollar amount or a percentage amount. How will your restaurant ask for tips? (For the case studies below, convert as appropriate by hand before entering the tip amount into your code.)
- Should the customer be able to see the input given by the waiter?
- Should the waiter be able to see the input given by the customer?
- Should the tip be calculated pre-tax or post-tax?
- How should information on the receipt be formatted? (*Hint: If you use the placeholder characters %6.2f in your fprintf function, you will be able to output a number with 2 decimal place precision. We'll talk more about what this placeholder is doing in our next class*).

Test Cases

Meal Price (CAD)	Tip Amount
17.75	12%
100.21	\$80
42.15	0

Note: If your restaurant location is outside of Canada, please provide a monetary conversion in your code, assuming that all inputs will remain in Canadian dollars. Remember to provide a reference for any conversion rates.

Reminders

- Variable names should be meaningful.
- Inputs and outputs should clearly communicate with the user.

Do Not Read Through to Part B until you have completed Part A

Part B

Problem Description

In testing out your code in the restaurant, you notice that some very angry customers have been entering negative values for their tips! Using the skills we've learned this week, rewrite your code to detect when the tip entered by the customers is negative and determine a reasonable way to modify that value so that your customers can't devalue their meal.

Provide a 3-4 sentence description of what happened in your original code when the angry customer left a negative tip, and what you chose to add to your code design to prevent this from happening.

Is this your ideal solution to solving the tipping problem? What would you change about your code, if you knew how to tell the computer to do so, and why would you make that change? Please provide a 1-2 sentence description.

Test Cases

Meal Price (CAD)	Tip Amount
30.21	-10.50

Hint: Relational operators produce logical outputs that can also be used in arithmetic calculations.

What you need to submit

Your code, command window display, and answers to the questions in this assignment should be submitted as a PDF file to Crowdmark.